



SOFTWARE ENGINEERING

C03001

CHAPTER 3 — REQUIREMENTS ENGINEERING

Anh Nguyen-Duc
Tho Quan Thanh

TOPICS COVERED

- ✓ Functional and non-functional requirements
- ✓ Requirements engineering processes
- ✓ Requirements elicitation
- ✓ Requirements specification
- ✓ Requirements validation
- ✓ Requirements change

TOPICS COVERED

- ✓ **Functional and non-functional requirements**
- ✓ Requirements engineering processes
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- ✓ Requirements specification
- ✓ Requirements validation
- ✓ Requirements change



FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS





How the customer explained it



How the project leader understood it



How the engineer designed it



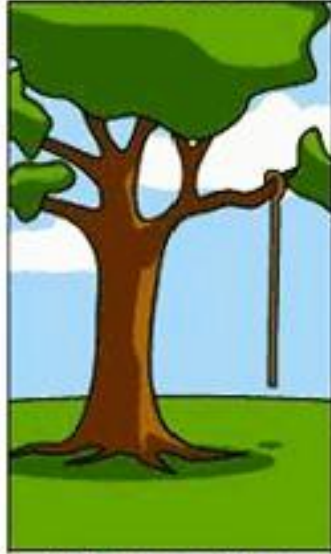
How the programmer wrote it



How the sales executive described it



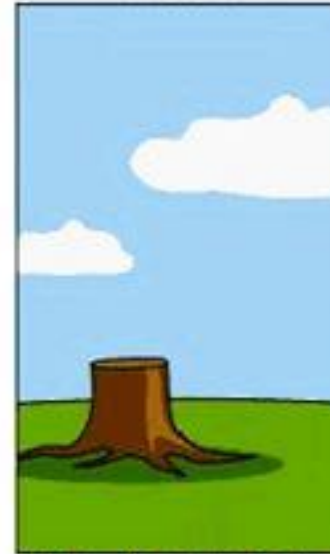
How the project was documented



What operations installed



How the customer was billed



How the helpdesk supported it



What the customer really needed

WHAT IS A REQUIREMENT?

Requirement engineering = establishing the **services** that the customer requires from a system and the **constraints** under which it operates and is developed.



- ✓ Requirement = the descriptions of
 - the system services
 - and constraints

- ✓ It may range
 - from a high-level abstract statement
 - to a detailed mathematical functional specification.

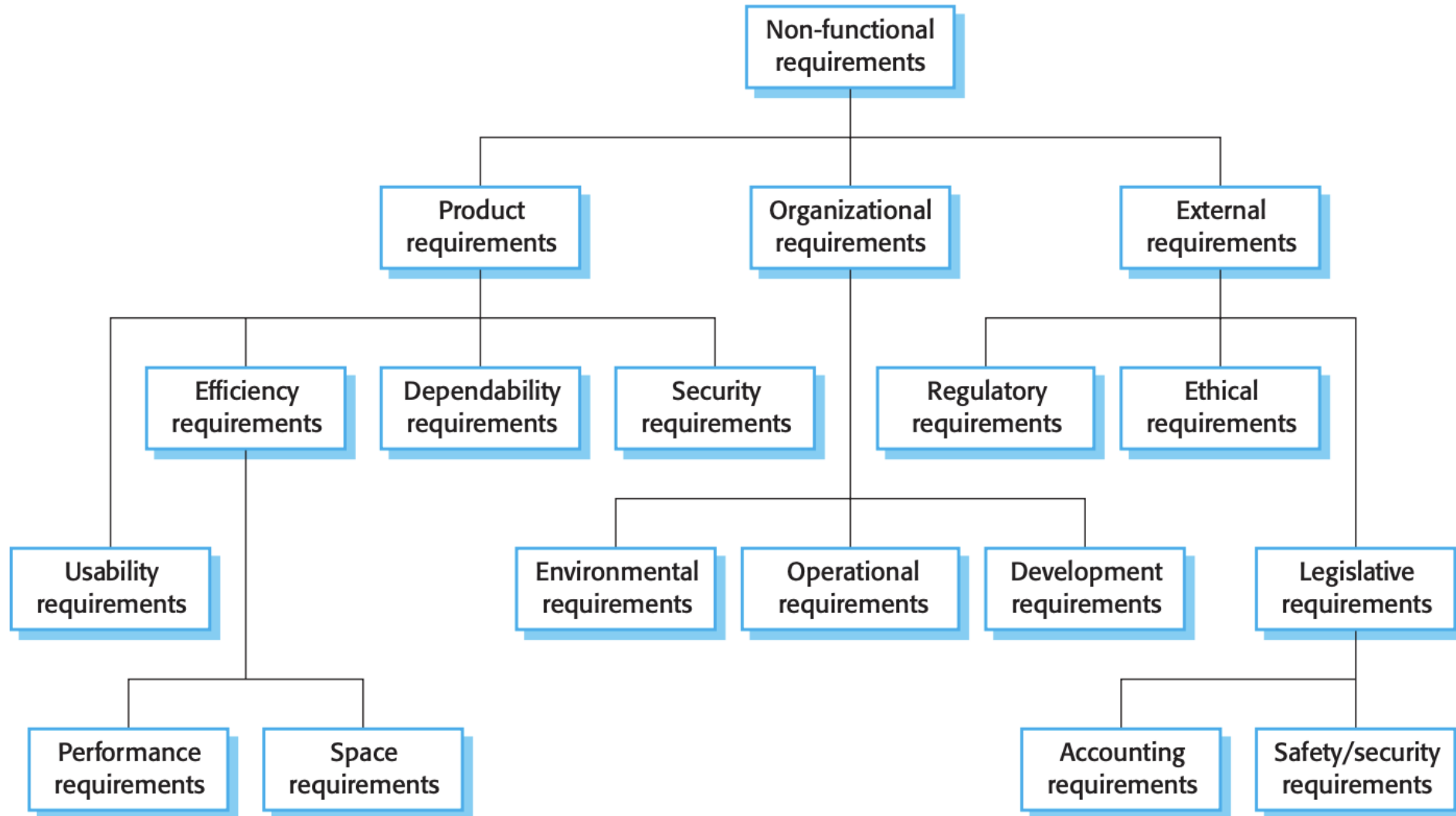
FUNCTIONAL REQUIREMENTS

- ✓ Statements of services the system should provide
 - ✓ How the system should react to particular inputs
 - ✓ How the system should behave in particular situations
 - ✓ State what the system should not do.
- ➔ Describe WHAT the system should do!

NON-FUNCTIONAL REQUIREMENTS

- ✓ Not directly concerned with the specific services delivered by the system to its users.
 - ✓ Usually specify or constrain characteristics of the system as a whole.
- ➔ Describe HOW WELL the system can conform!

NON-FUNCTIONAL REQUIREMENTS



GOALS VS. REQUIREMENTS

✓ Goal

- A general intention of the user.
- Usually high-level and describe the desired end result of a project
- Often less volatile

✓ Requirements are often

- Concrete
- Measureable
- Testable

GOALS VS. REQUIREMENTS (CONT.)

Goal:

The system should be easy to use by medical staff.



Non-functional requirement:

Medical staff shall be able to use all the system functions after four hours of training.

NON-FUNCTIONAL REQUIREMENTS

- ✓ Define system properties and constraints
 - Properties: reliability, response time and storage requirements.
 - Constraints: I/O device capability, system representations, etc.
- ✓ Non-functional requirements may be more critical than functional requirements.
 - If these are not met, the system may be useless.

SYSTEM STAKEHOLDERS

- ✓ Any person or organization who is affected by the system in some way and so who has a legitimate interest
- ✓ Stakeholder types
 - End users
 - System managers
 - System owners
 - External stakeholders

TOPICS COVERED

- ✓ Functional and non-functional requirements
- ✓ **Requirements engineering processes**
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- ✓ Requirements specification
- ✓ Requirements validation
- ✓ Requirements change



REQUIREMENTS ENGINEERING PROCESSES



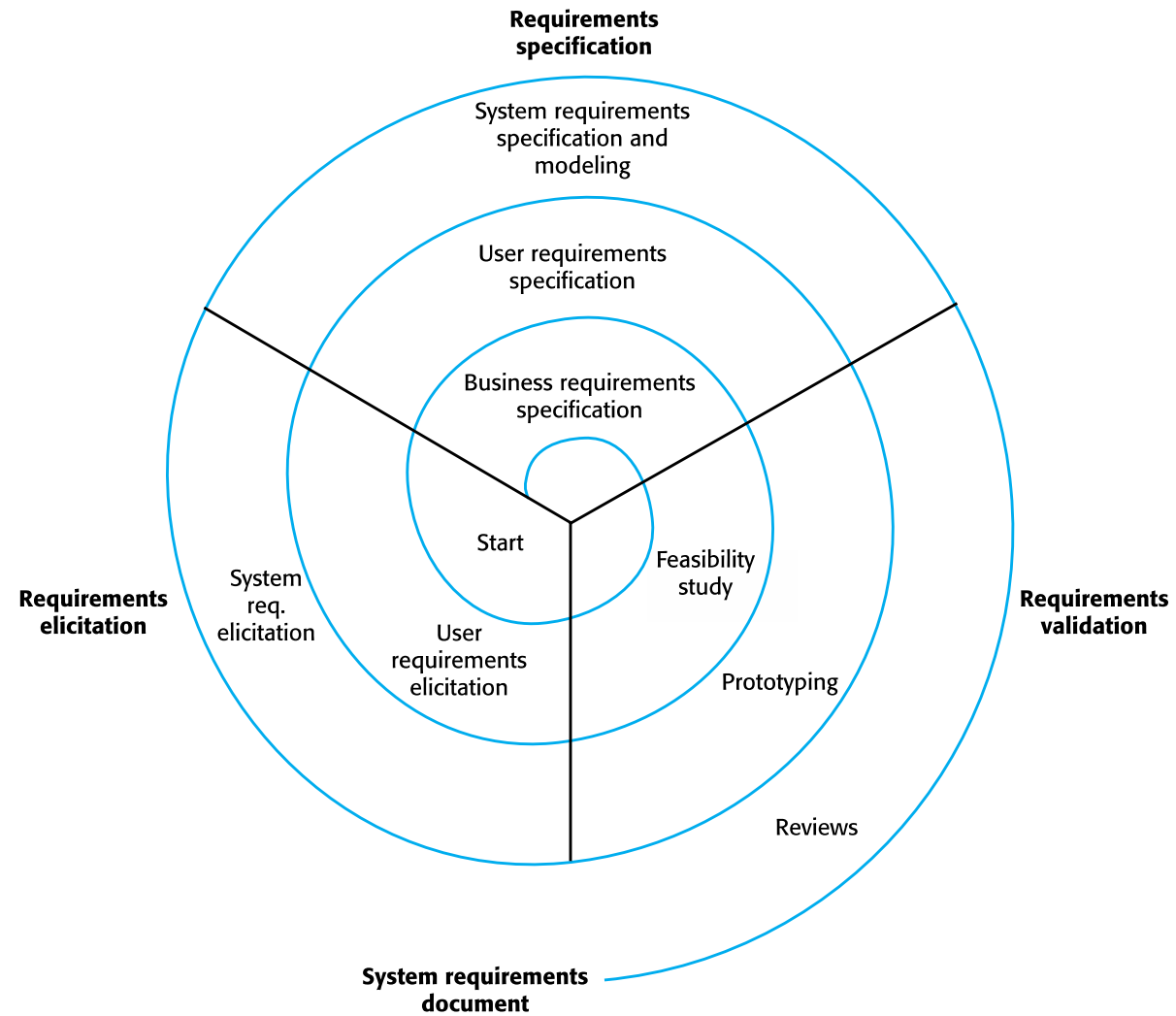
REQUIREMENTS ENGINEERING PROCESSES

- ✓ Processes to “generate” all requirements
- ✓ Generic activities common to all processes
 - Requirements elicitation;
 - Requirements specification;
 - Requirements validation.
- ✓ In practice, RE is an iterative activity



"The client kept changing the requirements on a daily basis, so we decided to freeze them until the next release."

A SPIRAL VIEW OF THE PROCESS



TOPICS COVERED

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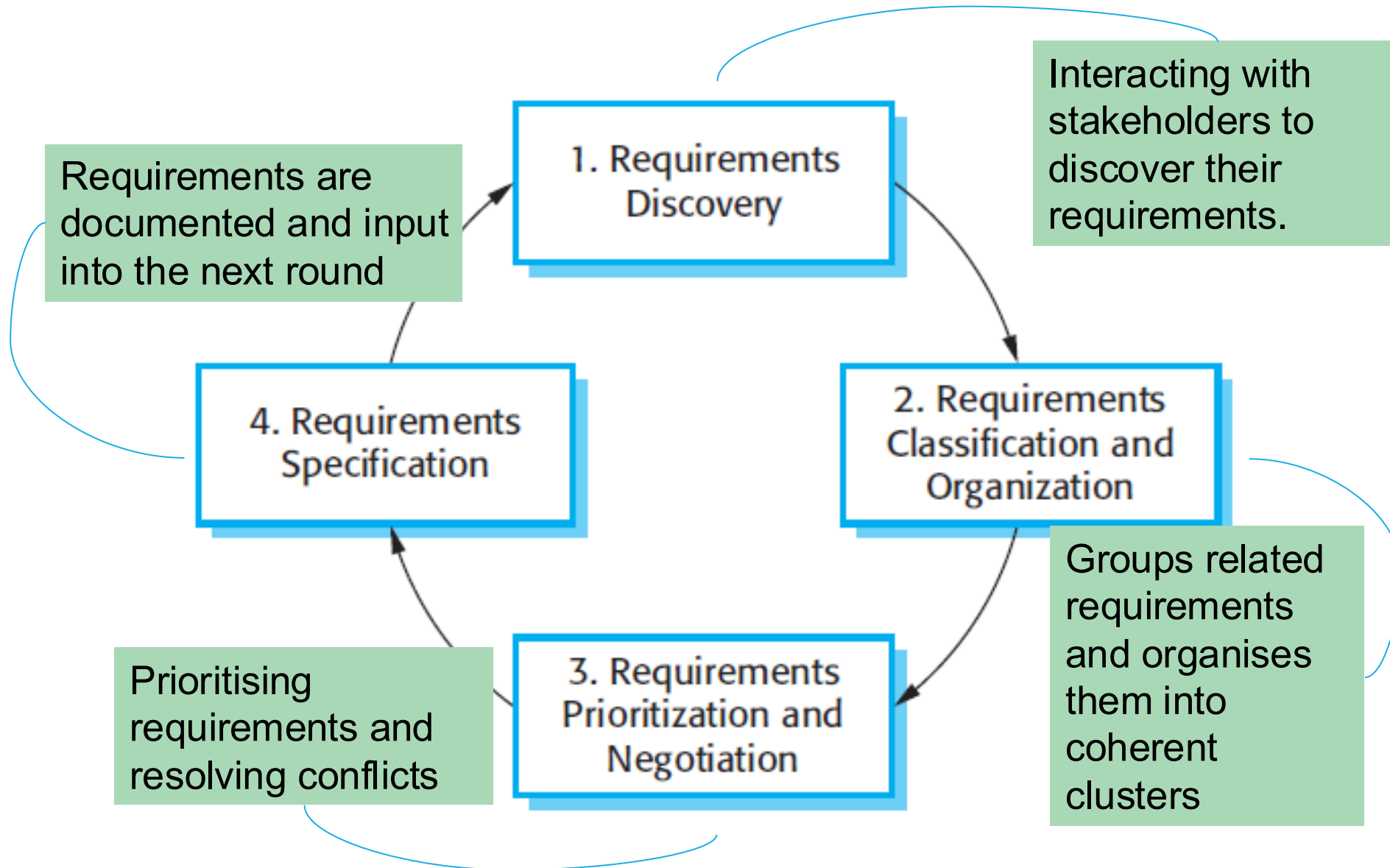
REQUIREMENTS ELICITATION AND ANALYSIS

- ✓ ~ requirements elicitation or requirements discovery.
- ✓ Work with customers to find out:
 - the application domain, the services and the operational constraints (system performance, hardware constraints, etc.).
- ✓ May involve
 - end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called stakeholders.

PROBLEMS OF REQUIREMENTS ELICITATION

- ✓ Stakeholders don't know what they really want.
- ✓ Stakeholders express requirements in their own terms.
- ✓ Different stakeholders may have conflicting requirements.
- ✓ Organisational and political factors may influence the system requirements.
- ✓ The requirements change during the analysis process.
 - New stakeholders may emerge and the business environment change.

THE REQUIREMENTS ELICITATION AND ANALYSIS PROCESS



REQUIREMENTS DISCOVERY

- ✓ To gather information about the required and existing systems and distil the user and system requirements from this information.
- ✓ Main concerns:
 - Stakeholders
 - Discovery techniques/approaches/...

DISCOVERY TECHNIQUE - INTERVIEWING



- ✓ Part of most RE processes.
- ✓ Types of interview
 - Closed vs Open => mixed?
- ✓ Be effective
 - Be open-minded, avoid pre-conceived ideas about the requirements and are willing to listen to stakeholders.
 - Prompt the interviewee to get discussions going using a springboard question, a requirements proposal, or by working together on a prototype system.

DISCOVERY TECHNIQUE - ETHNOGRAPHY

- ✓ Observational technique
 - used to understand operational processes and help derive support requirements for these processes
- ✓ How
 - A social scientist spends a considerable time observing and analysing how people actually work.
 - People do not have to explain or articulate their work.
 - Social and organisational factors of importance may be observed.

TOPICS COVERED

- ✓ Functional and non-functional requirements
- ✓ Requirements engineering processes
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- ✓ **Requirements specification**
- ✓ Requirements validation
- ✓ Requirements change



REQUIREMENTS SPECIFICATION

REQUIREMENTS SPECIFICATION

- ✓ The process of writing down the user and system requirements in a requirements document.
- ✓ Notes:
 - User requirements have to be understandable by end-users and customers who do not have a technical background.
 - System requirements are more detailed requirements and may include more technical information.
 - The requirements may be part of a contract for the system development

WAYS OF WRITING A SYSTEM REQUIREMENTS SPECIFICATION

Notation	Description
Natural language	Sentences in natural language. Each sentence should express one requirement.
Structured natural language	Natural language statements on a standard form or template
Design description languages	Uses a language like a programming language, but with more abstract features
Graphical notations	Graphical models, supplemented by text annotations (best for functional requirements); UML use case and sequence diagrams are commonly used.
Mathematical specifications	Based on mathematical concepts such as finite-state machines or sets; Can reduce the ambiguity but hard to understand (and hard to check manually)

WAYS OF WRITING A SYSTEM REQUIREMENTS SPECIFICATION

Notation	Description
Natural language	Sentences in natural language. Each sentence should express one requirement.

Waste collection is often designated to an organization that provides professional waste management services. A typical waste collection process involves (1) back officers, who operate a central system to create calendar, coordinate front collectors and janitors ,(2) collectors, who drive different types of vehicles and (3) janitors who manually collect garbage from Major Collecting Points (MCPs). Calendar and tasks were assigned among teams of janitors and coordinated by back officers. These assignments are often arranged in a weekly basic. Back officers also plan which vehicles to use and their routes. This planning activity happens every month. Everyday, the back officers sent messages with information about collecting route and time to collectors and janitors. Janitors use trollers (see Figure 1b) to collect garbage in their assigned areas and deliver to the MCPs. Collectors will pick up garbage from all janitors at an MCP. One collector drives only one vehicle during his working shift. The collector will drive through several MCPs with a predetermined route by back officers.

WAYS OF WRITING A SYSTEM REQUIREMENTS SPECIFICATION

Notation	Description
Natural language	Sentences in natural language. Each sentence should express one requirement.

- ✓ Used for writing requirements because it is expressive, intuitive and universal.
 - The requirements can be understood by users and customers.
- ✓ Problems
 - Lack of clarity: Precision is difficult without making the document difficult to read.
 - Requirements confusion: Functional and non-functional requirements tend to be mixed-up.
 - Requirements amalgamation: Several different requirements may be expressed together.

STRUCTURED SPECIFICATIONS

Notation	Description
Structured natural language	Natural language statements on a standard form or template

- ✓ Writing on a standard form or template:
 - Name
 - Inputs, outputs
 - The information needed for the computation
 - Action
 - Pre and post conditions (if appropriate)
 - The side effects (if any)
- ✓ Writing in a boiler plate template

Insulin Pump/Control Software/SRS/3.3.2

Function	Compute insulin dose: safe sugar level
Description	Computes the dose of insulin to be delivered when the current measured sugar level is in the safe zone between 3 and 7 units.
Inputs	Current sugar reading (r2); the previous two readings (r0 and r1).
Source	Current sugar reading from sensor. Other readings from memory.
Outputs	CompDose—the dose in insulin to be delivered
Destination	Main control loop.
Action	CompDose is zero if the sugar level is stable or falling or if the level is increasing but the rate of increase is decreasing. If the level is increasing and the rate of increase is increasing, then CompDose is computed by dividing the difference between the current sugar level and the previous level by 4 and rounding the result. If the result, is rounded to zero then CompDose is set to the minimum dose that can be delivered.
Requirements	Two previous readings so that the rate of change of sugar level can be computed
Pre-condition	The insulin reservoir contains at least the maximum allowed single dose of insulin.
Post-condition	r0 is replaced by r1 then r1 is replaced by r2.
Side effects	None

TABULAR SPECIFICATION

- ✓ Particularly useful when you have to define a number of possible alternative courses of action.
- ✓ Example:

Condition	Action
Sugar level falling ($r2 < r1$)	CompDose = 0
Sugar level stable ($r2 = r1$)	CompDose = 0
Sugar level increasing and rate of increase decreasing ($(r2 - r1) < (r1 - r0)$)	CompDose = 0
Sugar level increasing and rate of increase stable or increasing ($(r2 - r1) \geq (r1 - r0)$)	CompDose = round $((r2 - r1)/4)$ If rounded result = 0 then CompDose = MinimumDose

USE-CASE DIAGRAM

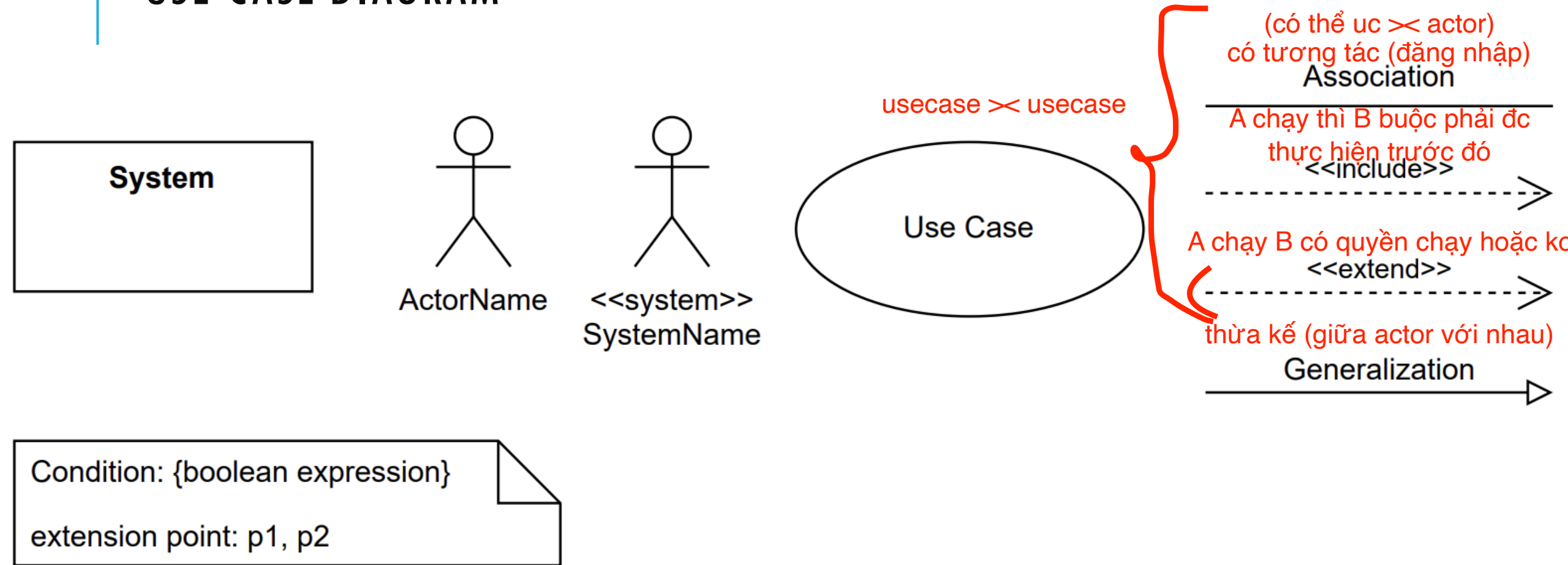


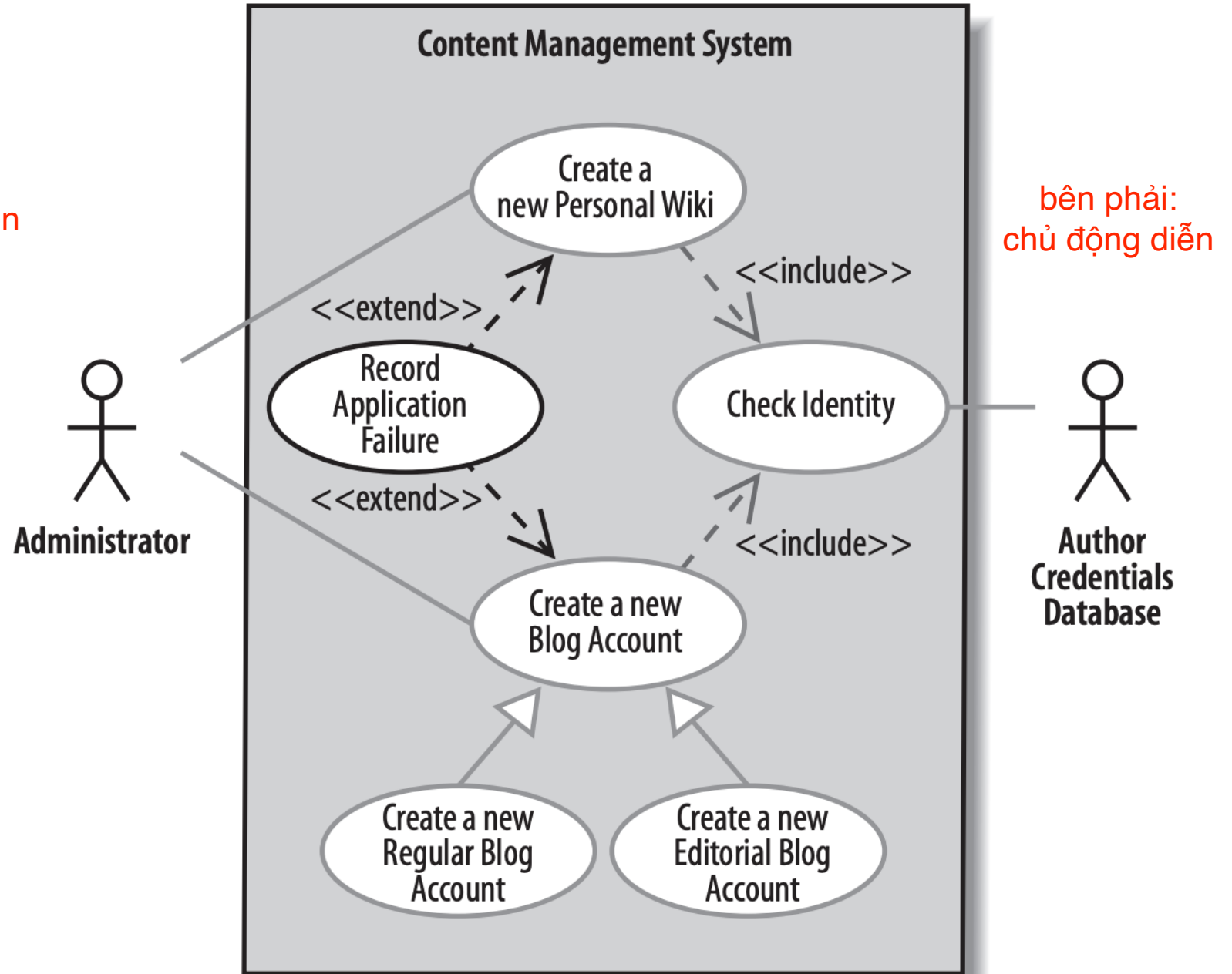
Figure 1: Main components of a use-case diagram

USE-CASE DIAGRAM

extend và include ngược chiều mũi tên

Figure 2: Simple use-case diagram for CMS

content management system

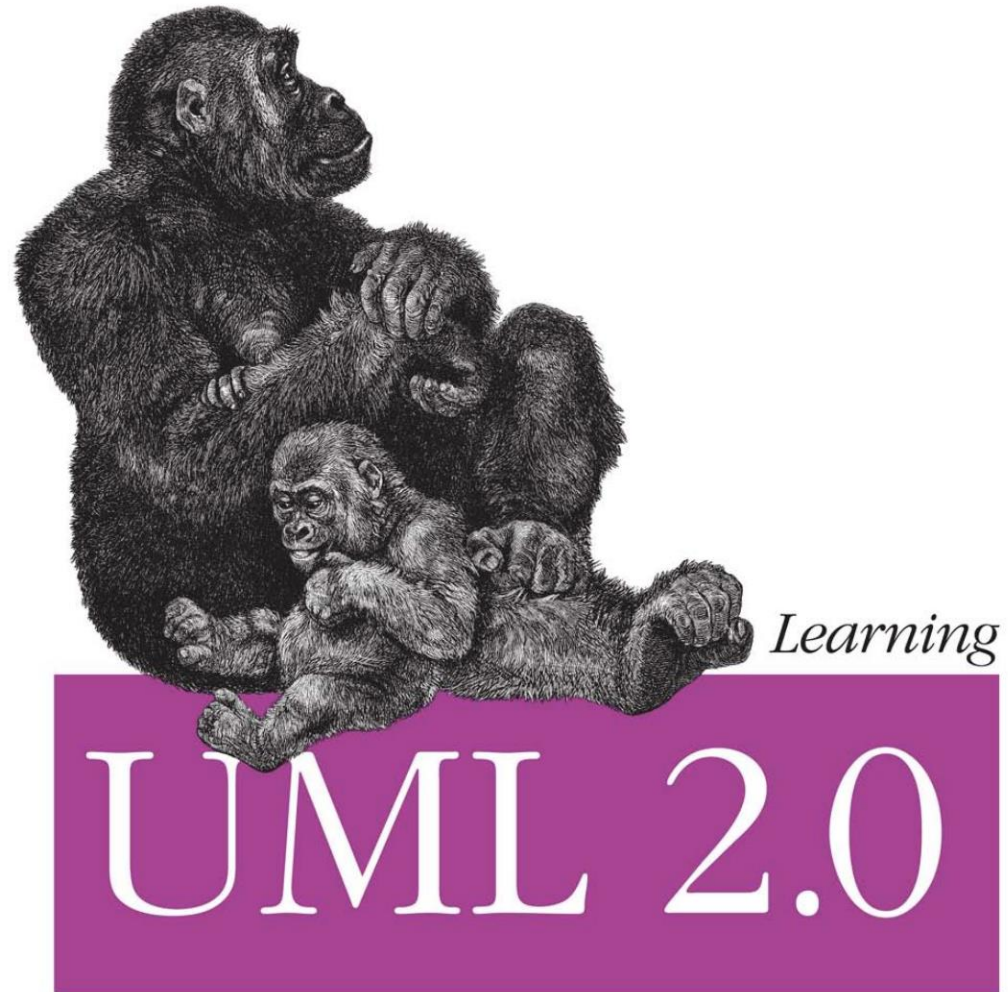


bên phải:
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USE-CASE DESCRIPTION

Figure 3: Use-case description table for “Create a new Blog Account”

Use case name	Create a new Blog Account	
Related Requirements	Requirement A.1.	
Goal In Context	A new or existing author requests a new blog account from the Administrator.	
Preconditions	The author has appropriate <u>proof of identity</u> .	
Successful End Condition	A new blog account is created for the author.	
Failed End Condition	The application for a new blog account is rejected.	
Primary Actors	Administrator.	
Secondary Actors	None .	
Trigger	The Administrator asks the CMS to create a new blog account.	
Included Cases	Check Identity	
Main Flow	Step	Action
	1	The Administrator asks the system to create a new blog account.
	2	The Administrator selects an account type.
	3	The Administrator enters the author's details.
	4	The author's details are checked.
	include::Check Identity	
	5	The new account is created.
	6	A summary of the new blog account's details are emailed to the author.
Extensions	Step	Branching Action
	4.1	The author is not allowed to create a new blog.
	4.2	The blog account application is rejected.
	4.3	The application rejection is recorded as part of the author's history.



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COMMON MISTAKES



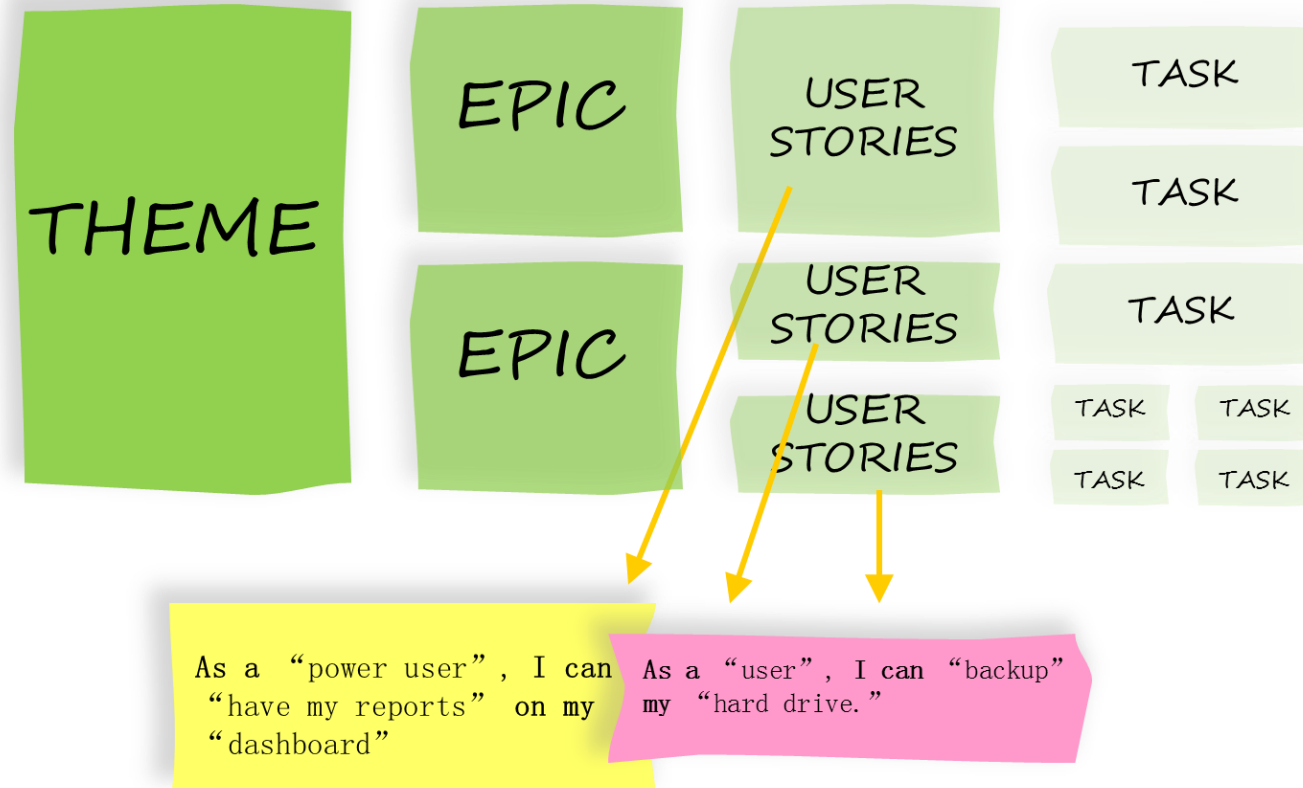
AGILE METHODS AND REQUIREMENTS

- ✓ Many Agile methods argue that producing detailed system requirements is a waste of time as requirements change so quickly.
- ✓ The requirements document is therefore always out of date.
- ✓ Agile methods usually use incremental requirements engineering and may express requirements as ‘user stories’
- ✓ This is practical for business systems but problematic for systems that require pre-delivery analysis (e.g. critical systems) or systems developed by several teams.

USER STORY

As a <type of user>, I want <some goal> so that <some reason>

USER STORIES



EXAMPLE

"As a returning customer, I want to log in with my fingerprint so that I can access my account quickly without typing a password."

- ✓ **FR-1:** The system **must prompt** the user for biometrics if a fingerprint template exists.
- ✓ **FR-2:** The system **must encrypt** the biometric data token during transit.
- ✓ **FR-3:** The system **must redirect** the user to the dashboard upon successful authentication.
- ✓ **FR-4:** The system **must display** an "Authentication Failed" error after three unsuccessful attempts.
- ✓ **FR-5:** The system **must fall back** to standard password entry if the biometric scanner fails.

TOPICS COVERED

- ✓ Functional and non-functional requirements
- ✓ Requirements engineering processes
- ✓ Requirements elicitation
- ✓ Requirements specification
- ✓ **Requirements validation**
- ✓ Requirements change



REQUIREMENTS VALIDATION

REQUIREMENTS CHECKING

- ✓ **Validity.**
 - Does the system provide the functions which best support the customer's needs?
- ✓ **Consistency.**
 - Are there any requirements conflicts?
- ✓ **Completeness.**
 - Are all functions required by the customer included?
- ✓ **Realism.**
 - Can the requirements be implemented given available budget and technology
- ✓ **Verifiability.**
 - Can the requirements be checked?

REQUIREMENTS VALIDATION TECHNIQUES

- ✓ Requirements reviews
 - Systematic manual analysis of the requirements.
- ✓ Prototyping
 - Using an executable model of the system to check requirements.
- ✓ Test-case generation
 - Developing tests for requirements to check testability.

TOPICS COVERED

- ✓ Course's project
- ✓ Functional and non-functional requirements
- ✓ Requirements engineering processes
- ✓ Requirements elicitation
- ✓ Quality of requirements
- ✓ Requirements specification
- ✓ Requirement in Agile
- ✓ Requirements validation
- ✓ **Requirements change**



REQUIREMENTS CHANGE

CHANGING REQUIREMENTS

- ✓ The business and technical environment of the system always changes after installation.
- ✓ The people who pay for a system and the users of that system are rarely the same people.
- ✓ Large systems usually have a diverse user community, with many users having different requirements and priorities that may be conflicting or contradictory.

REQUIREMENTS MANAGEMENT

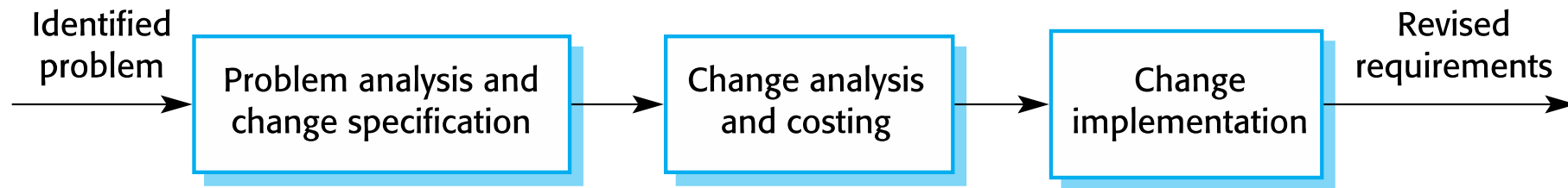
- ✓ Requirements management is the process of managing changing requirements during the requirements engineering process and system development.
- ✓ New requirements emerge as a system is being developed and after it has gone into use.
- ✓ You need to keep track of individual requirements and maintain links between dependent requirements so that you can assess the impact of requirements changes. You need to establish a formal process for making change proposals and linking these to system requirements.

REQUIREMENTS MANAGEMENT PLANNING

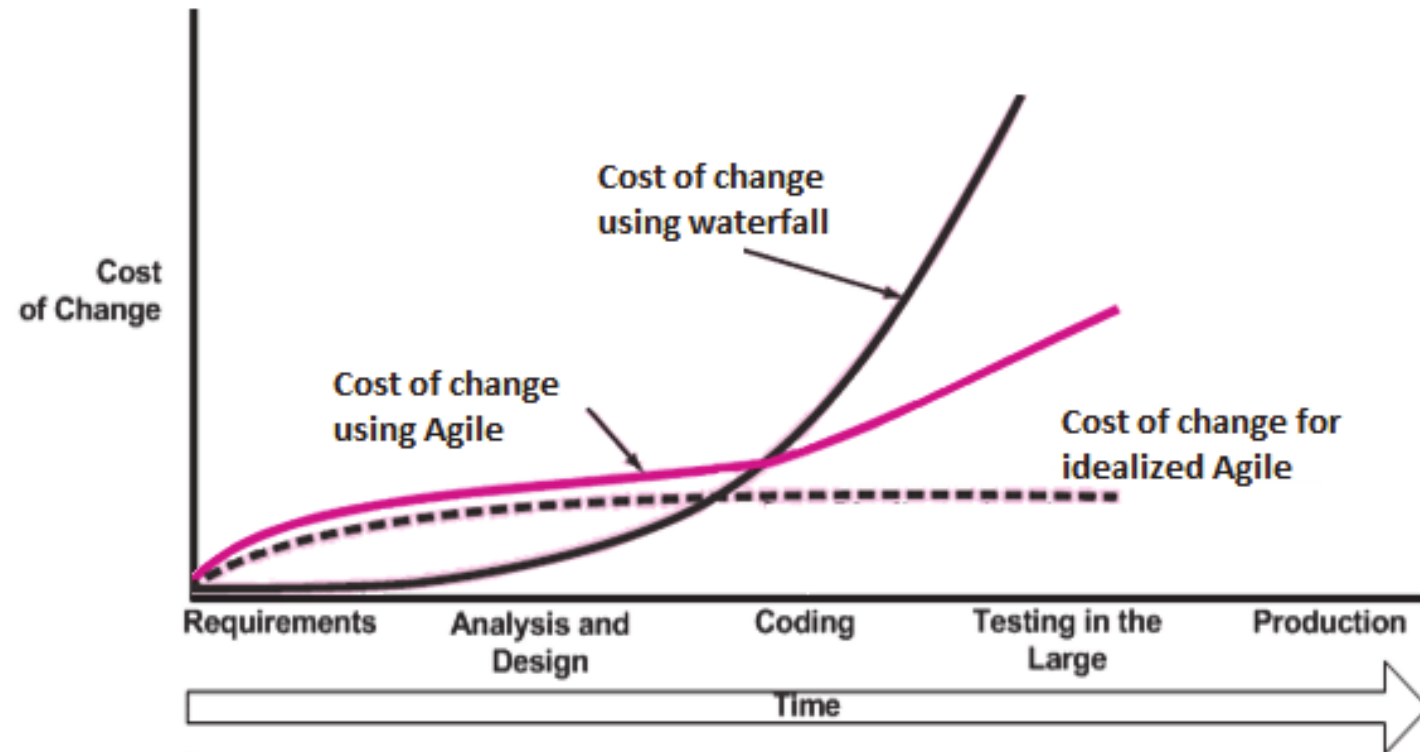
- ✓ Establishes the level of requirements management detail that is required.
- ✓ Requirements management decisions:
 - Requirements identification
 - A change management process
 - Traceability policies
 - Tool support

REQUIREMENTS CHANGE MANAGEMENT

- ✓ Deciding if a requirements change should be accepted
 - Problem analysis and change specification
 - Change analysis and costing
 - Change implementation



RESPONSE TO CHANGE: AGILE APPROACH!



SUMMARY

- ✓ Requirements: what the system should do and constraints on its operation and implementation.
- ✓ Functional requirements = the services
- ✓ Non-functional requirements = constraints (development & use)
 - apply to the system as a whole.
- ✓ The software requirements document (i.e. SRS) is an agreed statement of the system requirements.
- ✓ The RE process is an iterative process
 - requirements elicitation, specification and validation.

SUMMARY (CONT.)

- ✓ Requirements elicitation and analysis = iterative process
 - requirements discovery, classification and organization, negotiation and requirements documentation.
- ✓ Techniques for requirements elicitation
 - interviews, scenarios, use-cases and ethnography, etc.
- ✓ Requirements validation = checking the requirements
 - for validity, consistency, completeness, realism and verifiability.
- ✓ Business, organizational and technical changes inevitably
 - => changes to the requirements for a software system.
- ✓ Requirements management = managing and controlling the requirement changes.